IVAD / PROJECT SHARING SHEET

Exploring PCB Anomaly Detection with CNN and YOLO

A Hands-on Computer Vision PoC

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I built a small PoC to understand visual anomaly detection: prepare public VisA PCB data, explore normal-only CNN modelling, improve and stress-test preprocessing, and train supervised YOLO models. These are complementary explorations on separate board datasets.

Three technical explorations

A CNN Baseline

PCB1 / normal-only fitting

- Inspect the data
- Correct orientation & align
- WideResNet-50-2 features
- Per-position Gaussian / Mahalanobis distance
- Anomaly score & heatmap

OBSERVED

Detected 72/100 defects (72% recall). Orientation, registration and background interference mattered.

LEARNED

Inspect where the anomaly score comes from before adding complexity.

B Improve & Stress-Test

PCB2 / PCB3 - separate experiments

- Start from the CNN baseline
- Score inside the board ROI
- Calibrate the threshold
- Test lighting correction
- Evaluate robustness & decide

OBSERVED

PCB2: false alarms 13% to 3%; recall stayed 80%. PCB3: some lighting gains, worse false alarms under blur; correction not adopted.

LEARNED

A gain in one condition can create a new failure mode.

C Supervised YOLO

PCB4 / defect-labelled training

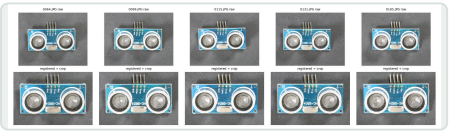
- Inspect defect annotations
- Convert masks to boxes
- Train YOLOv8s / YOLO11s
- Calibrate image detection
- Analyse failures & confounds

OBSERVED

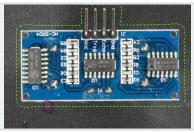
Image recall: YOLOv8s 98%, YOLO11s 95%. One generic defect class. Models were actually trained.

LEARNED

Higher recall here came with brightness confounding, unseen-type generalization limits and imprecise localization.



PCB1 / raw inputs above, aligned below



PCB2 / normal image, false alarm



PCB4 / label audit, not predictions

> Exploratory comparison: normal-only CNN fitting and supervised YOLO require different data and labels. The rounds use separate datasets; recall values do not establish architecture superiority. Image-level detection is not proof of accurate defect localization.

What I learned

- Data quality and preprocessing matter.
- Better metrics can introduce unexpected trade-offs.
- Investigate the root cause before increasing complexity.
- Choose models around available data, labels and real objectives.

Project & technical record

Website not yet deployed. Source access requires repository permission.

GitHub source (private; permission required): github.com/leonlaiyc/industrial-visual-anomaly-detection

Frozen score records, protocols, splits, audit and repair chronology are retained in the source. After PCB3 evaluation, an analysis input check was repaired (7dbd302); scores, thresholds and decisions were unchanged. D5 accepted by project owner: 2026-10-11. Overall portfolio acceptance is separate.

Data: VisA, Amazon Science, Zou et al., ECCV 2022, CC BY 4.0. Images modified by alignment, cropping, resizing and overlays. AI-assisted development. No validation on real manufacturing-line images; no production-readiness claim. No endorsement implied. Dataset source 繚 CC BY 4.0